

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – TRANSFORMER MECHANICS WORKSHEET

Course Code:	CSA65	Course Title:	Generative AI and Large Language Models
CO Assessed:	CO1 – Precision (BL1, BL2)	Assessment:	Assessment Tool 3 – Transformer Mechanics Worksheet
Weightage:	40% of CIA	Total Marks:	100 (1 Question)
CO1 Statement:	<i>Explain the fundamentals of Artificial Intelligence, Generative AI, Foundation Models, Transformer architecture, tokenization, embeddings, and Large Language Models. (BL1, BL2)</i>		
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Section / Batch:	CSE – AI	Date:	30/07/26

Instructions to Students

- Answer one questions. It carries 100 Marks. Total: 100 Marks.
- Tokenize a sentence.
- Convert tokens to token IDs.
- Explain embeddings.
- Label a Transformer encoder.
- Identify Query, Key, and Value.
- Explain which words receive higher attention.
- Compare Transformer models.
- Complete a Transformer pipeline.
- Draw a block diagram.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

SECTION A – TRANSFORMER MECHANICS WORKSHEET

26. Foundation Models Concept Worksheet

1. Tokenize a sentence

Sentence:

The cats are sleeping on the mat.

Tokens:

["The", "cats", "are", "sleeping", "on", "the", "mat", "."]

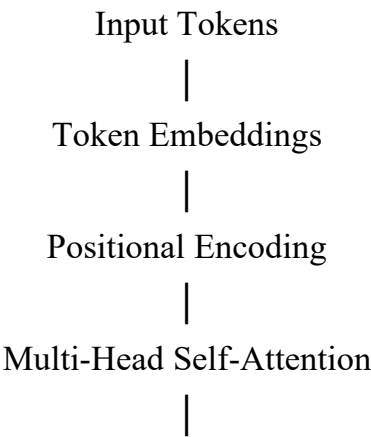
2. Convert tokens to token IDs

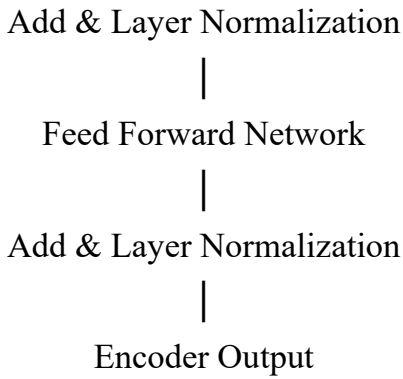
Token	Token ID (Example)
The	101
cats	2456
are	2024
sleeping	5777
on	2006
the	1996
mat	13523
.	102

3. Explain embeddings

Embeddings are dense numerical vectors that represent the meaning of words. Words with similar meanings have similar embedding vectors, helping the Transformer understand relationships between words.

4. Label a Transformer Encoder





5. Identify Query (Q), Key (K), and Value (V)

- **Query (Q):** Represents what the current token is searching for.
 - **Key (K):** Represents the information each token provides.
 - **Value (V):** Contains the actual information of each token.
-

Attention Formula:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

6. Explain which words receive higher attention

Sentence:

"The cats are sleeping on the mat."

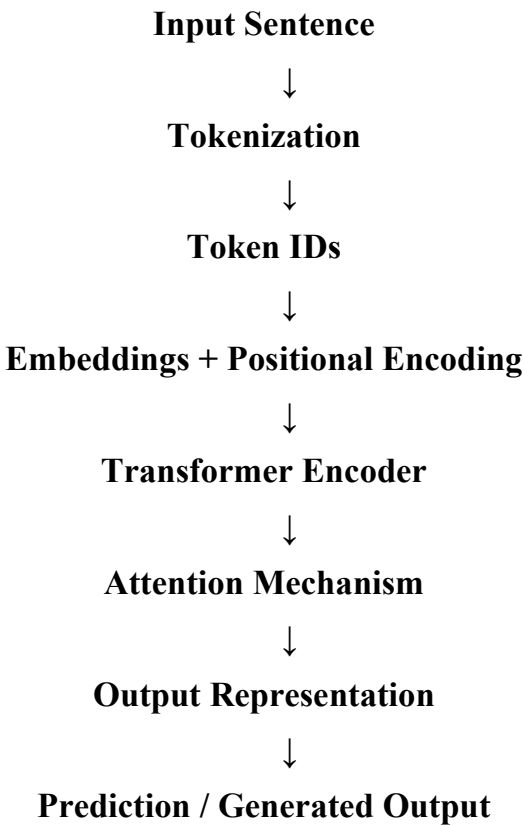
For the word "sleeping":

- **cats** → High attention (subject performing the action)
 - **mat** → Moderate attention (location related to the action)
 - **The, are, on** → Lower attention (function words with less semantic meaning)
-

7. Compare Transformer models

Model	Architecture	Direction	Typical Use
BERT	Encoder-only	Bidirectional	Classification, NER, embeddings
GPT	Decoder-only	Left-to-right	Text generation, chatbots
T5	Encoder-Decoder	Bidirectional encoder + autoregressive decoder	Translation, summarization, text-to-text tasks

8. Complete a Transformer pipeline



9. Block Diagram



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Mechanics Explanation	30%	All key concepts (AI, ML, DL, GenAI, LLMs) correctly identified with clear hierarchical relationships	Most concepts correct; minor gaps in relationships	Several concepts missing or misclassified	Major conceptual errors; map incomplete
Worked-Example Accuracy	25%	Real-world examples linked accurately to each concept	Examples mostly relevant	Few or generic examples	No real-world linkage shown
Mathematical Reasoning	20%	Logical, well-organized structure with clear connectors	Mostly organized; minor structural issues	Some disorganization affecting clarity	Poorly structured, hard to follow
Diagram & Working Clarity	15%	Visually clear, creative, easy to interpret	Neat and readable	Cluttered or inconsistent formatting	Untidy, difficult to interpret
Step Justification	10%	Concise labels/notes that add clarity	Adequate labeling	Minimal or unclear labeling	No supporting labels or explanation

Marks Evaluation:

Question No.	Technical / Concept(30%)	Application(25%)	Quality (20%)	Presentation (15%)	Documentation / Communication (10%)	Total Marks (100)
Q1						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____